

## Ideation Phase

### Brainstorm & Idea Prioritization Template

|               |                                                                              |
|---------------|------------------------------------------------------------------------------|
| Date          | 7 July 2026                                                                  |
| Team ID       | <b>SWTID-2026-2886</b>                                                       |
| Project Name  | Plugging into the Future: An Exploration of Electricity Consumption Patterns |
| Maximum Marks | 4 Marks                                                                      |

#### Brainstorm & Idea Prioritization Template:

Brainstorming provides a free and open environment that encourages everyone within a team to participate in the creative thinking process that leads to problem solving. Prioritizing volume over value, out-of-the-box ideas are welcome and built upon, and all participants are encouraged to collaborate, helping each other develop a rich amount of creative solutions.

### Step-1: Team Gathering, Collaboration and Select the Problem Statement

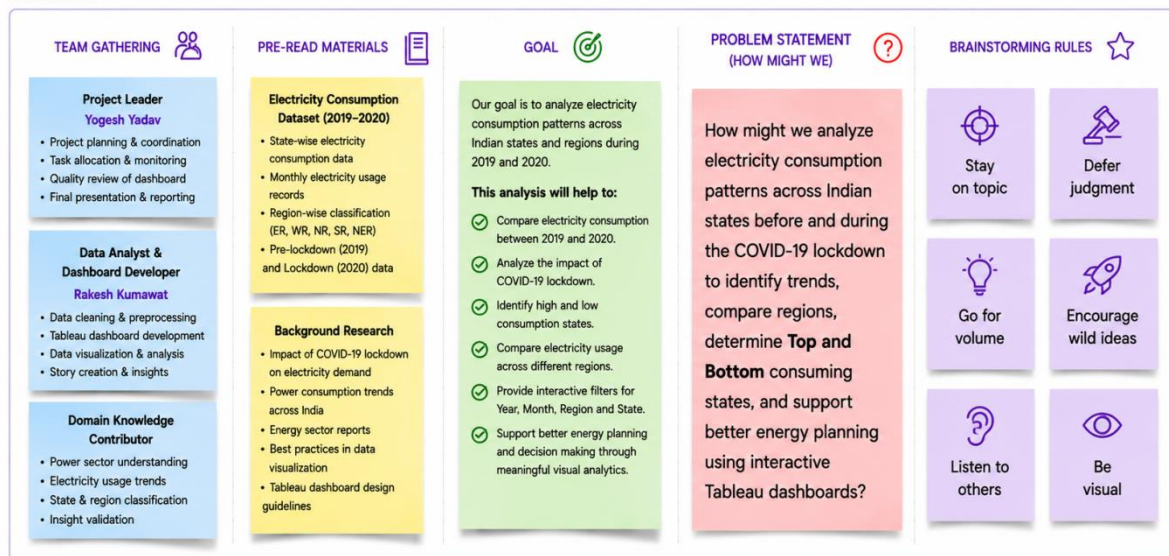
This is the core focus of the brainstorm, framed as a "How Might We" statement:

**How might we analyze electricity consumption patterns across Indian states before and during the COVID-19 lockdown to identify trends and support better energy planning using interactive Tableau dashboards?**

#### Key Rules of Brainstorming

- Stay focused on the project objective.
- Encourage creative ideas from every team member.
- Consider multiple visualization techniques.
- Prioritize ideas based on feasibility and impact.
- Validate every visualization using the dataset.
- Build an interactive dashboard for better analysis.

## STEP 1 Team Gathering, Collaboration & Select the Problem Statement



## Step-2: Brainstorm, Idea Listing and Grouping

After discussing different analytical approaches, the team grouped similar ideas into three major categories.

### Cluster 1: Interactive Dashboard & Visualization

The team proposed developing an interactive Tableau dashboard that allows users to explore electricity consumption through multiple visualizations.

Ideas included:

- Develop separate geographical maps for 2019 and 2020 electricity consumption.
- Create interactive bar charts to compare state-wise electricity usage.
- Design regional comparison charts to identify differences among Indian regions.
- Build a responsive dashboard combining all worksheets in a single interface.
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### Cluster 2: Dynamic User Interaction

The second discussion focused on improving user interaction and dashboard flexibility.

Ideas included:

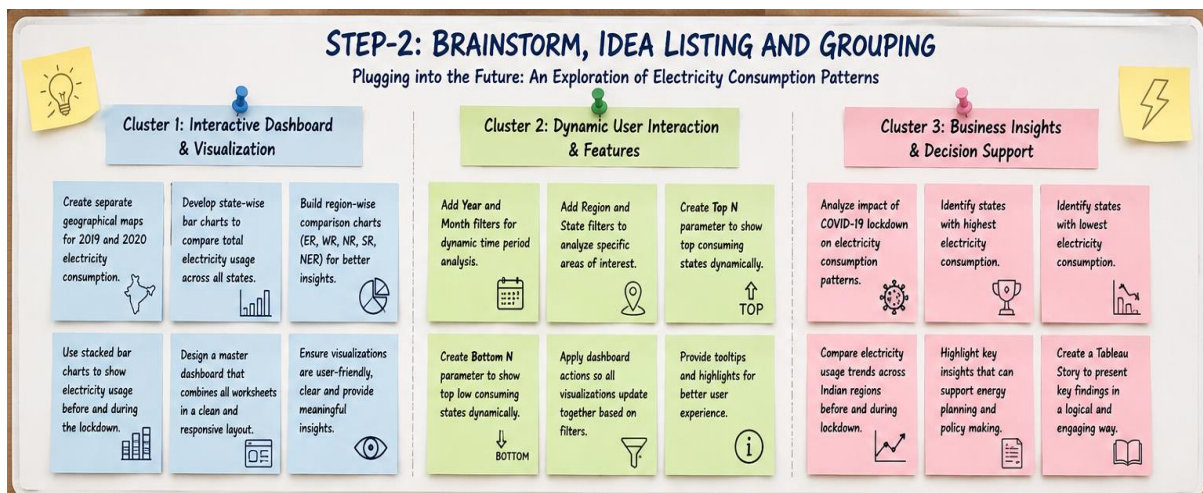
- Add Year, Month, Region, and State filters for dynamic exploration.
- Create Top N and Bottom N parameter-based visualizations.
- Allow dashboard filters to update all worksheets simultaneously.
- Enable users to compare electricity usage across different time periods.

### Cluster 3: Business Insights & Decision Support

The third group focused on generating meaningful business insights.

Ideas included:

- Analyze electricity demand before and during the COVID-19 lockdown.
- Identify states with the highest and lowest electricity consumption.
- Compare electricity usage across Eastern, Western, Northern, Southern, and North-Eastern regions.
- Create a Tableau Story to explain key findings in a logical sequence.



### Step-3: Idea Prioritization

After evaluating every idea based on **Importance** and **Feasibility**, the team selected the following priorities.

#### High Importance / High Feasibility (Top Priorities)

- Develop interactive Tableau dashboards with multiple visualizations.
- Build separate geographical maps for 2019 and 2020.
- Implement dynamic Year, Month, Region, and State filters.
- Create Top N and Bottom N parameter-based analysis.
- Design regional comparison charts for better understanding of electricity consumption.

#### High Importance / Low Feasibility (Strategic Long-Term Ideas)

- Integrate live electricity consumption data through APIs.
- Build AI-based electricity demand prediction models.
- Add real-time forecasting and anomaly detection.

- Develop automated electricity consumption alerts for policymakers.
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## Low Importance / High Feasibility (Quick Wins)

- Improve dashboard color themes and formatting.
- Add informative tooltips to every visualization.
- Include dashboard legends and descriptive titles.
- Enhance navigation using Tableau Story Points.

## Low Importance / Low Feasibility (Discarded Ideas)

- Develop a mobile application for dashboard access during the first phase.
- Integrate IoT-based real-time smart meter data.
- Build 3D visualization dashboards.
- Implement voice-controlled dashboard interaction.

